Data Analytics Portfolio

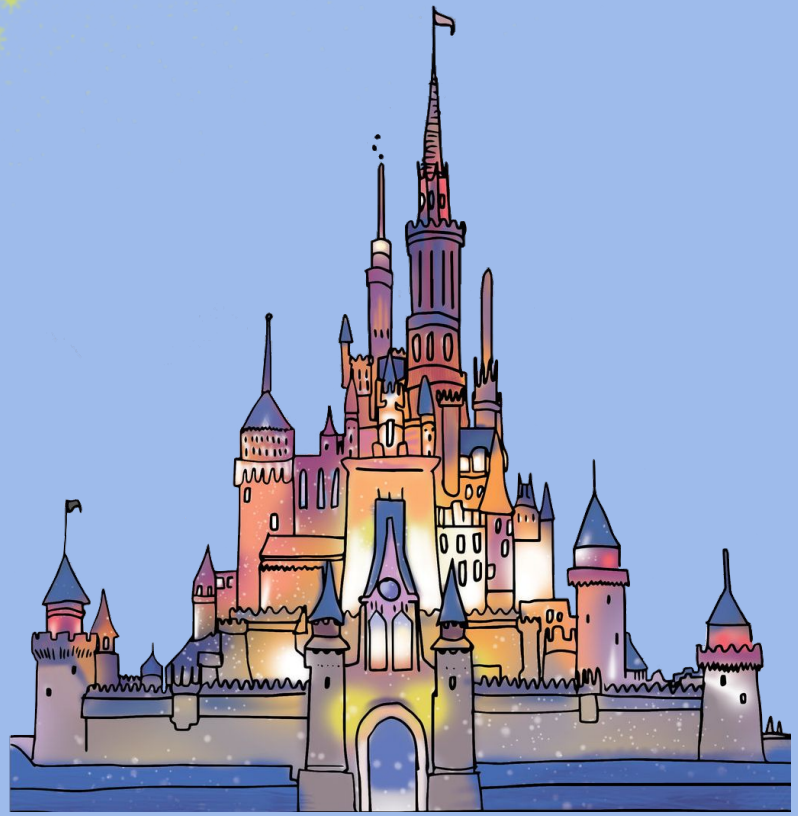
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Agenda

- Key Skills & Tools
- Tableau Revenue Optimization Dashboard
- Power BI Disney+ Content Analytics Dashboard
- Excel Fast Food Franchise Sales Analysis
- Automated Python + Excel Reporting
- Customer Churn Python Analysis
- Python Credit Score Prediction
- Contact

Key Skills & Tools

Technical Proficiency

Expertise in Python, SQL, Excel, Power BI, Tableau, and Power Query enables efficient data manipulation and visualization.

Data Analytics Skills

Skilled in data cleaning, transformation, dashboard creation, and visualization techniques for actionable insights.

AI and Machine Learning

Experience applying AI/ML models (scikit-learn) to solve complex problems such as prediction, classification, and automation.

Analytical Leadership Skills

Strong analytical problem-solving abilities combined with clear communication and teamwork skills for cross-functional projects.

Effective Communication

Ability to clearly present complex data insights to both technical and non-technical stakeholders.

Collaboration & Teamwork

Proven teamwork skills working across departments to deliver projects and shared goals.

Tableau Revenue Optimization

Objective

Analyze sales data from 349 video game stores to uncover patterns and identify top-performing products and locations to improve overall store performance.

Key Insights

Top stores and products were identified, revealing high revenue concentration in select locations and popular game titles driving sales.

Impact

Guided revenue optimization decisions that focused marketing and inventory efforts on high-performing stores and product lines, boosting profitability.

Dashboard Link

Explore the interactive Tableau dashboard to dive deeper into store-level revenue trends and performance metrics.

Tableau Dashboard

Power BI Disney+ Content Analytics Dashboard



Power BI Disney+ Content Analytics Dashboard

Project Objective

Analyze Disney+ content performance data to identify trends in viewer engagement and understand which shows and movies drive the most user interaction.

Key Features

Interactive filled map visualizes regional engagement, complemented by dynamic charts tracking content popularity over time and across genres.

Business Impact

Empowered content teams with actionable insights, enabling data-driven decisions to tailor programming and marketing strategies that increase viewer retention and satisfaction.

Explore Dashboard

Access the interactive Power BI dashboard to explore detailed content analytics and uncover new opportunities to delight Disney+ subscribers.

Power BI Dashboard

Excel Fast Food Franchise Sales Analysis

Project Link:

Excel Analysis

Total Sales & Average Ticket

Total sales reached 820,385 with an average ticket value of 3,131.24 across 262 order lines, highlighting strong transaction volume and value.

Sales Momentum & Trends

Overall sales increased by 14.2% over two months, showing positive momentum. Recent successful promotions and pricing strategies should be continued.

Top Products & Markets

Burgers led sales with 391,652 revenue. Lisbon was the top market, generating 283,185 in sales, suggesting targeted marketing and inventory focus here.

Channel & Payment Insights

In-store sales accounted for 40.4%, online 37.2%, and drive-thru 22.4%. Credit card was the dominant payment method, highlighting payment preferences.

Manager Impact & Recommendations

Top manager Joao Silva achieved 225,074 in sales. Documenting best practices here can support network-wide improvements and operational efficiency.

Excel Fast Food Franchise Sales Analysis

Raw Data

Product	Price	Quantity	Purchase Type	Payment Method	Manager
Fries	3.49	573.07	Online	Gift Card	Tom Jackson
Beverages	2.95	745.76	Online	Gift Card	Pablo Perez
Sides & Other	4.99	200.40	In-store	Gift Card	Joao Silva
Burgers	12.99	569.67	In-store	Credit Card	Walter Muller
Chicken Sandv	9.95	201.01	In-store	Credit Card	Walter Muller
Fries	3.49	573.07	In-store	Credit Card	Remy Monet
Fries	3.49	573.07	In-store	Credit Card	Remy Monet
Sides & Other	4.99	200.40	In-store	Credit Card	Walter Muller
Burgers	12.99	554.27	In-store	Credit Card	Remy Monet

Clean Data

Order ID	Date	Product	Price	Purchase Type	Payment Method	Manager2	Quantity	Revenue	City	Country/region
10452	11/7/2022	Fries	3.49	Online	Gift Card	Tom Jackson	574	2003.26	London	United Kingdom
10453	11/7/2022	Beverages	2.95	Online	Gift Card	Pablo Perez	746	2200.7	Madrid	Spain
10454	11/7/2022	Sides & Other	4.99	In-store	Gift Card	Joao Silva	201	1002.99	Lisbon	Portugal
10455	11/8/2022	Burgers	12.99	In-store	Credit Card	Walter Muller	570	7404.3	Berlin	Germany
10456	11/8/2022	Chicken Sandwiches	9.95	In-store	Credit Card	Walter Muller	202	2009.9	Berlin	Germany
10457	11/8/2022	Fries	3.49	In-store	Credit Card	Remy Monet	574	2003.26	Paris	France
10459	11/8/2022	Sides & Other	4.99	In-store	Credit Card	Walter Muller	201	1002.99	Berlin	Germany
10460	11/9/2022	Burgers	12.99	In-store	Credit Card	Remy Monet	555	7209.45	Paris	France
10461	11/9/2022	Chicken Sandwiches	9.95	In-store	Credit Card	Remy Monet	202	2009.9	Paris	France

Analysis

Data Analysis Questions

1) What is our best selling product?

2) What is our total revenue?

3) What is the revenue breakdown by payment method?

4) What is the Total Revenue of each Country/Region?

5) What product gives the most revenue?

6) What franchise (City) sells more online?

Row Labels	Quantity Sold
Beverages	35666
Fries	32067
Burgers	29572
Chicken Sandwiches	11184
Sides & Other	10050
Grand Total	**118539**

Sum of Revenue
$ 812,135.13

Row Labels	Sum of Revenue
Credit Card	49.72%
Cash	29.49%
Gift Card	20.79%
Grand Total	**100.00%**

Row Labels	Sum of Revenue
Portugal	283555.36
United Kingdom	211522.88
Spain	136430.04
Germany	100698.95
France	79927.9
Grand Total	**$ 812,135.13**

Row Labels	Sum of Revenue
Burgers	392000.28
Fries	125802.14
Beverages	125737.76
Chicken Sandwich	115139
Sides & Other	53455.95
Grand Total	**$ 812,135.13**

Sum of Revenue	Column Labels			
Row Labels	Drive-thru	In-store	Online	Grand Total
London	28649.37		182873.51	211522.88
Lisbon	20235.05	170592.69	92727.62	283555.36
Paris	35853.99	16228.96	27844.95	79927.9
Madrid	58682.03	75547.31	2200.7	136430.04
Berlin	35433.57	65265.38		100698.95
Grand Total	**$ 178,854.01**	**$ 327,634.34**	**$ 305,646.78**	**$ 812,135.13**

Findings

Select Manager ->	**Joao Silva**

Geography	
Country	Portugal
City	Lisbon

Revenue	
Fries	$ 47,717
Beverages	$ 56,130
Sides & Other	$ 19,354
Burgers	$ 123,562
Chicken Sandwiches	$ 36,793
Total	**$ 283,555**

Automation of Python + Excel Reporting

```python
import pandas as pd
import win32com.client as win32
# import view the database
sales_table = pd.read_excel('Vendas.xlsx')

# view the database
pd.set_option('display.max_columns', None)
print(sales_table)

# revenue by store
revenue = sales_table[['ID Loja', 'Valor Final']].groupby('ID Loja').sum()
print(revenue)

# number of products sold per store
quantity = sales_table[['ID Loja', 'Quantidade']].groupby('ID Loja').sum()
print(quantity)

print('-' * 50)

# average price per product in each store
average_price = (revenue['Valor Final'] / quantity['Quantidade']).to_frame()
# will transform in a table
average_price = average_price.rename(columns={0: 'Average Price'})
print(average_price)
# send an email with the report using Outlook
#import in line 2
```

```python
outlook = win32.Dispatch('outlook.application')
mail = outlook.CreateItem(0)
mail.To = 'sabrinavalevaz@gmail.com'
mail.Subject = 'Sales Report by Store'
mail.HTMLBody = f'''
<p>Hello team,</p>

<p>Follow the sales report for each store.</p>

<p>Revenue:</p>
{revenue.to_html(formatters={'Revenue': '${:,.2f}'.format})}

<p>Quantity Sold:</p>
{quantity.to_html()}

<p>Average price per product in each store:</p>
{average_price.to_html(formatters={'Average Price': '${:,.2f}'.format})}

<p>If you have any questions, please feel free to contact me.</p>

<p>Best,</p>
<p>Sabrina Vaz</p>
'''

mail.Send()

print('Email Sent')
```

Objective

Streamlined large-scale data processing and reporting by automating Excel report generation, cutting down manual effort and turnaround time drastically.

Impact

Reduced reporting time, enabling quicker insights delivery and freeing up valuable team time for strategic analysis.

Link to Code

Access the full Python automation scripts on GitHub to explore the implementation.

Automation Project

Customer Churn Python Analysis

```
# !pip install pandas numpy openpyxl ipykernel plotly nbformat


# Import database
import pandas as pd

tabela = pd.read_csv("cancelamentos.csv")

# View database
tabela = tabela.drop(columns="CustomerID")
#print(tabela)

# null and non-null values
#print(tabela.info())
tabela = tabela.dropna() #NaN not a number, null value, remove rows
print(tabela.info())

# Cancelation Analysis
print(tabela["cancelou"].value_counts()) # count the cancelation values

print(tabela["cancelou"].value_counts(normalize=True).map("{:.1%}".format)) # % of cancela

# Cancelation Cause Analysis
import plotly.express as px
# Create
for coluna in tabela.columns: # create one graph for each column

  grafico = px.histogram(tabela, x=coluna, color="cancelou")
  # show graph
  #grafico.show()

tabela = tabela[tabela["duracao_contrato"]!="Monthly"] # no monthly contract duration
tabela = tabela[tabela["ligacoes_callcenter"]<=4] # no calls to call center (over 4 calls)
tabela = tabela[tabela["dias_atraso"]<=20] # no payment delay

print(tabela["cancelou"].value_counts(normalize=True))
```

Objective

Analyze main causes of customer churn using historical data and predictive modeling to inform retention efforts.

Impact

Churn linked to service dissatisfaction and pricing; recommended targeted offers and improved communication to reduce churn.

Code Repository

Explore Python visualization and analysis scripts on GitHub.

Customer Churn Project

Python Credit Score Prediction

Project Link:

Prediction Code

Project Objective

Classify customers into Poor, Standard, or Good credit risk categories using a dataset of 100,000 rows and 25 features to support credit approval and limit decisions.

Model Performance

Random Forest achieved the best accuracy at approximately 82.3%, outperforming KNN at 73.7% on a holdout test split, demonstrating strong predictive power.

Business Actions

Poor credit leads to manual reviews and secured card offers; Standard gets calibrated approvals and incentives; Good customers receive instant approval and loyalty perks.

Impact Thinking

Used macro-F1 scores and classification to ensure reliable, fair decisions and minimize false approvals or declines.

Governance & Fairness

Built metrics tailored to different demographics, explained results clearly with compliant reasons, and kept performance stable over time.

Next Steps

Planning to put into production with pipelines, tuned models, balanced classes, business-driven thresholds, and workflows that are easy to reproduce.

Python Credit Score Prediction

```python
import pandas as pd

tabela = pd.read_csv("clientes.csv")
print(tabela)
print(tabela.info()) # show null values, data type...

# encode database columns

from sklearn.preprocessing import LabelEncoder

codificador = LabelEncoder()

tabela["profissao"] = codificador.fit_transform(tabela["profissao"])

tabela["mix_credito"] = codificador.fit_transform(tabela["mix_credito"])

tabela["comportamento_pagamento"] = codificador.fit_transform(tabela["comportamento_pagamento"])

# who do I want to predict, who do I want to use to make the prediction

y = tabela["score_credito"]
x = tabela.drop(columns=["score_credito", "id_cliente"]) # name of the columns I want to delete

# training (60%-80%) e test (20%-40%)

from sklearn.model_selection import train_test_split

x_treino, x_teste, y_treino, y_teste = train_test_split(x, y, test_size=0.3)
```

```python
#create A.I., Decision tree, KNN, Nearest Neighbors

from sklearn.ensemble import RandomForestClassifier
from sklearn.neighbors import KNeighborsClassifier

modelo_arvoredecisao = RandomForestClassifier()
modelo_knn = KNeighborsClassifier()

# training the A.I.

modelo_arvoredecisao.fit(x_treino, y_treino)
modelo_knn.fit(x_treino, y_treino)

# test models

previsao_arvoredecisao = modelo_arvoredecisao.predict(x_teste)
previsao_knn = modelo_knn.predict(x_teste)

from sklearn.metrics import accuracy_score

print(accuracy_score(y_teste, previsao_arvoredecisao))
print(accuracy_score(y_teste, previsao_knn))

# choose which model is better

tabela_nova = pd.read_csv("novos_clientes.csv")
print(tabela_nova)

tabela_nova["profissao"] = codificador.fit_transform(tabela_nova["profissao"])
tabela_nova["mix_credito"] = codificador.fit_transform(tabela_nova["mix_credito"])
tabela_nova["comportamento_pagamento"] = codificador.fit_transform(tabela_nova["comportamento_pagamento"])

previsoes = modelo_arvoredecisao.predict(tabela_nova)
print(previsoes)
```

Let’s Connect!

Thank you for taking the time to review my portfolio. I am excited about the opportunity to bring my data analytics skills and passion for storytelling to Disney.

Feel free to reach out via email, LinkedIn, or phone to connect or ask any questions. I look forward to collaborating and contributing to Disney's innovative, data-driven projects!



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